

Internet of Things

Lecture 3

Challenges of IoT

The biggest challenges for IoT adoption include:

- Security Challenges
- Regulation Challenges
- Compatibility Challenges
- Bandwidth Challenges
- Customer Expectation Challenges

Challenges of IoT

1. Security Challenges

The rapid development of technology and the increasing complexity of cyberattacks have increased the risk of security breaches. IoT devices can be easy targets for hackers, so software developers must design secure applications to protect devices and users' data.

Example: A hacker accesses a smart camera without permission.

Key idea: Protect IoT devices and data from cyberattacks.

Challenges of IoT

2. Regulation Challenges

The IoT industry is relatively new, and specific regulations may be lacking in some areas. Regulations are needed to ensure that manufacturers follow appropriate security and privacy requirements. GDPR is an example of a regulation that protects personal data and privacy.

Example: A company collects personal data from smart devices without following applicable data protection requirements.

Key idea: Establish and follow rules for security and privacy.

Challenges of IoT

3. Compatibility Challenges

IoT devices must connect and communicate with each other to exchange data. However, different manufacturers may use different communication protocols and technical standards, making communication difficult.

Example: A smart bulb from one manufacturer cannot communicate with a smart home hub from another manufacturer.

Key idea: Ensure that different IoT devices can work together.

Challenges of IoT

4. Bandwidth Challenges

A large number of connected devices generate and transmit data, which can consume network bandwidth and cause congestion, delays, or reduced performance.

Example: Many smart cameras transmit video simultaneously, slowing down the network.

Key idea: Manage network capacity to support large numbers of connected devices.

Challenges of IoT

5. Customer Expectation Challenges

Customers expect IoT devices to be easy to use, reliable, affordable, secure, and responsive. If devices fail to meet these expectations, customers may not adopt or continue using them.

Example: Customers may stop using a smart thermostat if it frequently disconnects or responds slowly.

IoT Networking Considerations & Challenges

- Range
- Bandwidth
- Power usage
- Intermittent connectivity
- Interoperability
- Security

IoT Networking Considerations & Challenges

Range

- Networks can be described in terms of the distances over which data is typically transmitted by the IoT devices attached to the network: PAN – LAN –MAN-WAN
- **PAN(Personal Area Network**) is short-range, where distances can be measured in meters.

Example: a wearable fitness tracker device that communicates with an app on a cell phone over BLE.

- **LAN(Local Area Network)** is short- to medium-range, where distances can be up to hundreds of meters
- **Example:** Factory sensors communicate with a gateway through Wi-Fi

IoT Networking Considerations & Challenges

Range

- **MAN (Metropolitan Area Network)** is long-range (city wide), where distances are measured up to a few kilometers.
- **Example:** smart parking sensors installed throughout a city that are connected in a mesh network topology.
- **WAN (Wide Area Network)** is long-range, where distances can be measured in kilometers.
- **Example**: Agricultural sensors monitor temperature, humidity, and other environmental conditions across a large farm.

IoT Networking Considerations & Challenges

- An IoT network must collect data from IoT devices and transmit it to the correct destination. Therefore, we must choose a communication protocol that supports the required communication distance, or range.
- **For example**, BLE is designed for short-range communication, so it is generally unsuitable for a WAN application that needs to communicate over several kilometers.
- If transmitting data over long distances is difficult, edge computing can help. **Edge computing** processes and analyzes data close to the IoT devices instead of sending all raw data to a distant data center. This can **reduce communication requirements, latency, and network traffic**.

IoT Networking Considerations & Challenges

Bandwidth

- Bandwidth is the amount of data that can be transmitted over a network per unit of time.
- It determines how quickly data can be collected from IoT devices and transmitted to other devices, gateways, or cloud servers.
- **Bandwidth** is commonly measured in bits per second (bps), such as Kbps, Mbps, and Gbps.

IoT Networking Considerations & Challenges

Bandwidth is affected by many factors, which include:

1. Volume of data generated by each device
2. Number of devices deployed
3. Continuous transmission versus intermittent bursts: IoT devices may transmit data **continuously** or send data occasionally in **bursts**.

- **Continuous transmission**: Devices send data constantly, creating a relatively steady demand for bandwidth.
- **Intermittent bursts:** Devices send data occasionally, producing sudden increases in network traffic.
- **Peak periods:** Times when many devices transmit simultaneously, potentially causing network congestion.

IoT Networking Considerations & Challenges

Explain four techniques that can reduce bandwidth requirements in IoT networks, and discuss their limitations.

- **Packets that are too large.** If a device sends only a few bytes, a large packet may contain unused space filled with padding, wasting network capacity. **Packets that are too small.** Splitting a large amount of data into many small packets creates additional overhead because each packet needs its own header and other control information.
- **Data transmission rates are asymmetrical** (Upload and download speeds are not always equal)
- **Wireless and cellular networks** are traditionally low bandwidth, so consider whether a wireless technology is the right choice for high-volume applications.

IoT Networking Considerations & Challenges

- **Sampling less** frequently that reduces the number of measurements but may miss rapid changes.
- **Filtering** removes insignificant data, but may discard useful information.
- **Aggregation** combines multiple readings into summaries, but reduces data granularity.
- **Burst transmission** sends stored data in batches, but may introduce delays.

IoT Networking Considerations & Challenges

Power

- **Power** usage refers to the electrical energy consumed by an IoT device when it operates, processes data, and transmits information through a network.
- It is an important consideration because many IoT devices operate using batteries and may be installed in remote locations where replacing or recharging batteries is difficult.

IoT Networking Considerations & Challenges

Explain methods for reducing power consumption in IoT devices.

You must consider:

- the power Source such as a battery, solar cell, or capacitor
- the device's Lifecycle how long the device can operate before its power source needs replacement or recharging,
- use sleep Mode whenever it is idle.

IoT Networking Considerations & Challenges

- **Intermittent connectivity**

It means that an IoT device does not maintain a continuous network connection. It may connect periodically, or it may lose its connection because of network problems.

Causes of Intermittent Connectivity

1. Devices connect periodically to connect only when they need to send or receive data.
2. A weak signal or unstable network may cause a device to disconnect unexpectedly.
3. multiple devices may try to transmit data using the same communication channel at the same time. This competition can cause delays and collisions

IoT Networking Considerations & Challenges

How Can IoT Systems Handle Intermittent Connectivity

1. Devices temporarily save collected data when the network is unavailable and transmit it when connectivity returns.
2. A system can switch to another available network or communication path when the primary connection fails.

IoT Networking Considerations & Challenges

- **Interoperability**

It is the ability of different devices, equipment, systems, and technologies to communicate, exchange data, and work together successfully.

- **Security**

Security is a priority. Selection of networking technologies that implement end-to-end security, including authentication, encryption, and open port protection is crucial.